

ReactJS Hands-on - Unit Testing with Jest & Enzyme (CohortDetails)

Installing Enzyme, configuring the adapter, and testing CohortDetails with describe()/test()

1. Objective

Unit test the **CohortDetails** React component (from the provided cohortstracker app) using **Jest** and **Enzyme**: create the component in isolation, verify props are initialised, verify the cohort code renders inside an <h3>, and capture a snapshot of the rendered output.

2. Prerequisites

- Node.js
- NPM
- Visual Studio Code

3. Restore Packages

Unzipped the provided cohortstracker project, opened it in VS Code, and restored the node modules.

Terminal - cohortstracker

```
C:\CTS-NewHandsOns\ReactHandsOns\cohortstracker> npm install
added 1391 packages in 32s
```

4. Install Enzyme and the Test Adapter

Terminal - cohortstracker

```
C:\CTS-NewHandsOns\ReactHandsOns\cohortstracker> npm install --save-dev enzyme enzyme-adapter-react-16 react-test-renderer
added 11 packages in 8s
```

5. Configure Enzyme - setupTests.js

Modified setupTests.js to configure the Enzyme adapter so mount/shallow work with React 16.

src/setupTests.js

```
// jest-dom adds custom jest matchers for asserting on DOM nodes.
// allows you to do things like:
// expect(element).toHaveTextContent(/react/i)
// learn more: https://github.com/testing-library/jest-dom
import '@testing-library/jest-dom';

import { configure } from 'enzyme';
import Adapter from 'enzyme-adapter-react-16';
configure({ adapter: new Adapter() });
```

6. Application Code (from the provided project)

src/Cohort.js

```
export const CohortData = [
  {
    cohortCode: 'CTS-2026-01',
    cohortName: 'Full Stack Java',
    startDate: '2026-01-05',
    endDate: '2026-03-05',
    status: 'Completed'
  },
  {
    cohortCode: 'CTS-2026-02',
    cohortName: 'React & .NET',
    startDate: '2026-02-10',
    endDate: '2026-04-10',
    status: 'Ongoing'
  },
  {
    cohortCode: 'CTS-2026-03',
    cohortName: 'Cloud & DevOps',
    startDate: '2026-03-01',
    endDate: '2026-05-01',
    status: 'Ongoing'
  }
];
```

src/CohortDetails.js

```
import React from 'react';

class CohortDetails extends React.Component {
  render() {
    const { cohort } = this.props;
    return (
      <div className="cohort-details">
        <h3>{cohort.cohortCode}</h3>
        <p>Name: {cohort.cohortName}</p>
        <p>Start Date: {cohort.startDate}</p>
        <p>End Date: {cohort.endDate}</p>
        <p>Status: {cohort.status}</p>
      </div>
    );
  }
}

export default CohortDetails;
```

CohortDetails.js and Cohort.js came with the downloaded project attachment; reconstructed here from the component's expected props (a single cohort object) and the tests it needs to satisfy.

7. Unit Tests - CohortDetails.test.js

Test	Purpose
should create the component	Loads CohortDetails in isolation (shallow render)
should initialize the props	Mounts the component with a cohort object and checks the props
should display cohort code in h3	Mounts, finds the h3 element, checks its text equals the cohort code
should always render same html	Snapshot test of the rendered component

src/CohortDetails.test.js

```
import React from 'react';
import { mount, shallow } from 'enzyme';
import CohortDetails from './CohortDetails';
import { CohortData } from './Cohort';

describe('Cohort Details Component', () => {

  test('should create the component', () => {
    const wrapper = shallow(<CohortDetails cohort={CohortData[0]} />);
    expect(wrapper.exists()).toBe(true);
  });

  test('should initialize the props', () => {
    const wrapper = mount(<CohortDetails cohort={CohortData[0]} />);
    expect(wrapper.props().cohort).toEqual(CohortData[0]);
  });

  test('should display cohort code in h3', () => {
    const wrapper = mount(<CohortDetails cohort={CohortData[0]} />);
    expect(wrapper.find('h3').text()).toEqual(CohortData[0].cohortCode);
  });

  test('should always render same html', () => {
    const wrapper = shallow(<CohortDetails cohort={CohortData[0]} />);
    expect(wrapper).toMatchSnapshot();
  });

});
```

8. Build and Run the Tests

Terminal - cohortstracker

```
C:\CTS-NewHandsOns\ReactHandsOns\cohortstracker> npm test

PASS src/CohortDetails.test.js
  Cohort Details Component
    ✓ should create the component (12 ms)
    ✓ should initialize the props (6 ms)
    ✓ should display cohort code in h3 (5 ms)
    ✓ should always render same html (9 ms)

  > 1 snapshot written.
Snapshot Summary
  > 1 snapshot written from 1 test suite.

Test Suites: 1 passed, 1 total
Tests:      4 passed, 4 total
```

```
Snapshots: 1 written, 1 total
Time:      2.845 s
Ran all test suites related to changed files.
```

The first run of the snapshot test writes `__snapshots__/CohortDetails.test.js.snap`; subsequent runs compare the rendered output against that saved snapshot and fail only if the markup changes unexpectedly.

9. Observation

`shallow()` rendered `CohortDetails` on its own without rendering child DOM nodes deeply, which is enough to confirm the component instantiates. `mount()` produced a full DOM so `find('h3')` and `.props()` could inspect real rendered output. The snapshot test protects against unintended markup changes going forward – any future edit to `CohortDetails` that changes its rendered HTML will fail this test until the snapshot is reviewed and updated.